



MUHAMMAD IRSYA ZAELANI

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Tangerang, Banten

Computer Science student with an interest in cybersecurity and machine learning, committed to rigor, precision, punctuality, communication, and professionalism

EDUCATION

IPB University - Bogor, Indonesia

August 2025 - Now

Bachelor of Computer Science (S1)

- Focus on data structures, algorithms, and Machine Learning.
- 2nd Place, IPB Excellent Challenge Competition (IEC)
- Relevant coursework: Programming, Discrete Mathematics, Statistics, and Calculus

SMAN 28 KABUPATEN TANGERANG

June 2022 - July 2025

Science Program (MIPA)

- Scientific Methodological Analysis: Focus on hypothesis testing, processing data from science experiments, and preparing systematic technical documentation
- Participant, National Science Olympiad (OSN) - Informatics
- Staff, Student Anti-Corruption Movement
- Chairman, Robotics Club: Led the club and taught C/Arduino Programming Fundamentals to members

EXPERIENCE

Event Division UX Today - IT Today IPB

April 2026 – Now

- Planned and structured detailed event rundowns and timelines to ensure the smooth execution of each session
- Coordinated closely with team members, speakers, and external vendors to ensure all event requirements were met on schedule
- Checked, tested, and prepared presentation equipment ahead of each session to minimize technical disruptions during the event

Member - IPB Archery

August 2023 – Now

- Planned and structured detailed event rundowns and timelines arrangements to support the smooth execution of each session
- Coordinated closely with team members, speakers, and external vendors to ensure all event requirements were met on schedule
- Checked, tested, and prepared presentation equipment ahead of each session to minimize technical disruptions during the event

Head of D'Astha Robotics Club - SMAN 28 Kabupaten Tangerang

June 2023 – August 2024

- Integrating the principles of physics and mathematical logic in the design of mechanical systems as well as robot programming for technical competitions
- Manage organizational project management, from the component research stage to the implementation of robotic design
- Led team coordination in systematically troubleshooting hardware and software issues, identifying root causes across sensor wiring, circuit connections, and code logic to ensure reliable robot performance during competitions

Staff of D'Astha Karate Club - SMAN 28 Kabupaten Tangerang

February 2023 – April 2024

- Participated in regular training sessions, including technique drills and kata (form) exercises to build strength, discipline, and body coordination
- Applied focus and perseverance in mastering movement sequences, reinforcing discipline both on and off the mat
- Practiced and refined fundamental karate techniques through consistent, structured training routines

ORGANIZATIONAL EXPERIENCE

Reaserch and Technology Division - Internship BEM SSMI **September 2026 – Now**

- Provided technical support in developing and maintaining digital tools, systems, and websites for organizational needs
- Communicated and coordinated with other divisions to support organizational activities

Member of D'Astha Reaserch Team - SMAN 28 Kabupaten Tangerang **February 2023 – April 2024**

- Compiled comprehensive research reports evaluating the effectiveness of technology-based solutions for environmental and social issues, and collaborated with the team to present findings in a logical, organized manner
- Participated in various academic research competitions, including scientific essay writing and other research-based contests, applying structured methodology from problem formulation to conclusion
- Designed and conducted a water purification experiment using sulfur-based filtration, including assembling the testing apparatus and presenting the findings to an academic audience
- Practiced and refined fundamental karate techniques through consistent, structured training routines

Staf of Student Anti-Corruption Movement - SMAN 28 Kabupaten Tangerang **June 2023 – June 2024**

- Participated in campaigns and educational activities aimed at raising awareness of anti-corruption values and encouraging students to understand the importance of integrity and ethical behavior
- Developed and prepared simple outreach materials to communicate anti-corruption messages effectively to classmates and members of the wider school community
- Collaborated with fellow team members to organize and support awareness activities, ensuring that campaign materials and activities were delivered effectively and on schedule

VOLUNTEER EXPERIENCE

Voulenteer LPK Tepi Sawah **February 2026 – Now**

- Designed and structured an educational curriculum, outlining teaching materials, topics, and learning objectives tailored to children's needs
- Conducted direct field visits to teach, mentor, and engage with children in person, ensuring the curriculum was delivered effectively
- Designed and structured an educational curriculum, outlining teaching materials, topics, and learning objectives tailored to children's needs
- Facilitated interactive play and games sessions to build rapport with children and reinforce learning in a fun, engaging way

Volunteer Arduino Programming Instructor **June 2023 – August 2024**

- Taught fundamental C and Arduino programming concepts to club members, breaking down technical concepts into accessible, hands-on lessons
- Guided beginner members through hardware assembly and basic circuit wiring, helping them apply physics and logic principles to real robot builds
- Mentored individual members in debugging simple code and hardware issues, fostering independent problem-solving skills for future competitions

PROJECTS

DETAK - Early Autism Triage via Pose Estimation

- Developed a machine learning-based screening tool that analyzes body movement patterns from video to help identify early signs of autism, designed to run on low-cost, low-spec devices
- Trained and evaluated the detection model using real-world behavioral datasets, applying statistical analysis to identify the most meaningful movement patterns
- Built with offline accessibility in mind, enabling use in remote or underserved areas with limited internet access and healthcare resources; also explored browser- and mobile-based versions for wider reach

Zae Labs - Personal Portfolio

- Designed and built a personal portfolio website using React, Vite, and Framer Motion for smooth animations and interactions
- Integrated an AI-powered chat assistant using Google's Gemini API, complete with text-to-speech voice output
- Optimized the site for mobile devices and implemented secure handling of API keys and credentials using environment variables

ACHIEVEMENTS & CERTIFICATIONS

Certifications

- IBM SkillsBuild - Generative AI (2025)
- Gemini Certified Educator
- Certificate of National Teacher Training Amplification Students Batch 3
- Code Generation and Optimization Using IBM Granite
- IBM Granite Models for Software Development

Achievements

- 2nd Place, IPB Excellent Challenge Competition (2025)
- Participant, National Science Olympiad (OSN) - Informatics

SKILLS

- **Tools:** Excel, Figma, Canva, Ms Word
- **Soft Skills:** Teamwork, communication, problem-solving, adaptability, time management
- **Language:** Bahasa Indonesia (Native), Bahasa English (Intermediate)